🔑 **Answer Key — Teacher Copy · Lesson 6-8**

MSTAR PRACTICE WORKSHEET

Lesson 6-8: Generate Equivalent Expressions

Grade 6 Mathematics · Standard 6.AT.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which property is shown? $5 + 13 = 13 + 5$

- **A** Commutative Property
- B Associative Property
- C Identity Property
- D Distributive Property

> The order of the addends changed, so this is the Commutative Property of Addition.

Question 2 SELECTED RESPONSE — CORRECT: C

Which property is shown? $(6 \times 3) \times 2 = 6 \times (3 \times 2)$

- A Commutative Property
- B Identity Property
- **C** Associative Property
- D Distributive Property

> The grouping (parentheses) changed but the order stayed the same, so this is the Associative Property of Multiplication.

Question 3 SELECTED RESPONSE — CORRECT: D

Which property is shown? $47 + 0 = 47$

- (A) Commutative Property
- (B) Associative Property
- (C) Distributive Property
- (D) Identity Property

> Adding 0 does not change the value, so this is the Identity Property of Addition.

Question 4 SELECTED RESPONSE — CORRECT: D

Which property lets you rearrange 8×5 to 5×8?

- (A) Associative Property
- (B) Identity Property
- (C) Distributive Property
- (D) Commutative Property

> Changing the order of factors is the Commutative Property of Multiplication.

Question 5 SELECTED RESPONSE — CORRECT: D

Track 2 on the playlist plays $5(7 + 2) = 5 \times 7 + 5 \times 2$. Which property is this?

- (A) Commutative Property
- (B) Associative Property
- (C) Identity Property
- (D) Distributive Property

> Multiplying 5 across both numbers inside the parentheses (5×7 and 5×2) shows the Distributive Property.

Question 6 SELECTED RESPONSE — CORRECT: C

Track 3 plays (7 + 3) + 5 = 7 + (3 + 5). Only the parentheses moved. Which property is this?

- A Commutative Property
- B Distributive Property
- C Associative Property
- D Identity Property

The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

Which property of operations is shown by the equation $(7 + 4) + 6 = 7 + (4 + 6)$?

(A) Commutative Property of Addition

(B) Identity Property of Addition

(C) Distributive Property

(D) Associative Property of Addition

> The addends stay in the order 7, 4, 6 on both sides. Only the grouping changes, which is the associative property. Both sides have a value of 17.

- **A:** Commutative changes the ORDER of the addends. Here 7, 4 and 6 stay in order and only the parentheses move.
- **B:** Identity adds zero and leaves a number unchanged. No zero appears in this equation.
- **C:** Distributive spreads a multiplication across a sum. This equation contains no multiplication.

Part B — correct answer: D

Which reasoning shows why your answer to Part A is correct?

(A) The addends appear in a different order on the two sides, which is the commutative property.

(B) Adding zero leaves a number unchanged, which is the identity property.

(C) A factor outside the parentheses is multiplied by each term inside, which is the distributive property.

(D) The addends appear in the same order on both sides, but the parentheses move. Changing the grouping without changing the sum is the associative property, and both sides equal 17.

> Grouping changed and order did not, so the equation shows the associative property of addition.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Regrouping a Subtraction

Jamal wrote (20 – 8) – 5 = 20 – (8 – 5). He said the associative property lets him move the parentheses anywhere, so both sides must have the same value of 7.

Explain Jamal's misconception and give the correct value of each side.

Model answer: The associative property holds for addition and multiplication only. Moving the parentheses in a subtraction changes the value: (20 – 8) – 5 = 12 – 5 = 7, but 20 – (8 – 5) = 20 – 3 = 17. The two sides are not equal, so the equation is false.

2 points	Student explains that the associative property applies to addition and multiplication but not to subtraction, and shows that the left side equals 7 while the right side equals 17.
1 point	Student computes both sides as 7 and 17 but does not explain that the associative property does not apply to subtraction.
0 points	Student agrees with Jamal that both sides equal 7 or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Explain why the commutative property works for addition and multiplication but NOT for subtraction and division. Give a specific number example for each operation to prove your point.

Model answer: Addition and multiplication combine two amounts into a total or a product, and swapping which one you name first does not change that total: 10 + 2 = 2 + 10 = 12 and 10 × 2 = 2 × 10 = 20. Subtraction and division ask 'how much is left' or 'how many fit,' which depends on which number starts: 10 – 2 = 8 but 2 – 10 = –8, and 10 ÷ 2 = 5 but 2 ÷ 10 = 0.2.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.